When Order and Time Disagree: Position Overrides Explicit Time in Language Models

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Abstract

When a context states several values of the same quantity—a variable rewritten in a chain of thought, a user's city across chat sessions, a setting across commits—a language model must decide which value is current. Deployed contexts often present such records out of chronological order: feeds and logs list the newest entry first, and retrieval memories rank passages by relevance, while every record still carries its date. We ask which of the two a model follows when they disagree, holding the content fixed and varying only the order. In shuffled chains of thought that overwrite a variable eight times, all 14 open models we tested answer with the last-presented value in 89.5–99.5% of items. On MemConflict, a third-party long-term memory benchmark, showing the same dated sessions newest first costs Qwen3 21–40 accuracy points (32B to 4B) and Gemma 3 47–53 points (12B and 27B); ordering them by retrieval relevance costs 10–19 points. Neither scale nor reasoning removes the effect: Gemma 3 is worse at 27B than at 12B, and Qwen3's thinking mode removes most of it in short conversations but only halves it in long sessions, leaving 10 points at 32B. In LongMemEval's official retrieval-and-reading pipeline, merely skipping its date sort lowers knowledge-update accuracy by 14–17 points (Qwen3-4B with thinking, Qwen3-8B with and without), and the loss falls on the questions whose newest evidence is ranked first. Order has no effect on PersonaMem or LoCoMo questions that do not require choosing between an older and a newer state. We locate two failure points: in short contexts the model represents which value is newer but answers by position and copies the presented order into its own reasoning; in long sessions, dates written in session headers are not attached to the facts they govern. A model trained to list the dated records of the queried attribute in its thinking stops depending on order when it writes the list, but across eight training variants, making it write the list exactly when a question needs it remained a trade-off.

1 Introduction

Much of what language models read is a record of change. A chain of thought (CoT) assigns a value and later revises it; a chat assistant's memory holds the user's address before and after a move; a log lists the successive values of a configuration key. To answer "what is the value now?", a reader must order these statements in time. In most training text this is trivial, because text is written in the order events happen: the statement written last is also the most recent. *Later in the text* is therefore an excellent proxy for *later in time*.

Many contexts that models read at deployment break this proxy. E-mail clients, `git log`, changelogs and feeds list the newest entry first. Retrieval-augmented memories return passages by relevance, not by date. Agents append retrieved memories after their own trajectory; monitors read other models' reasoning out of order; tools concatenate outputs from many sources. In all of these,

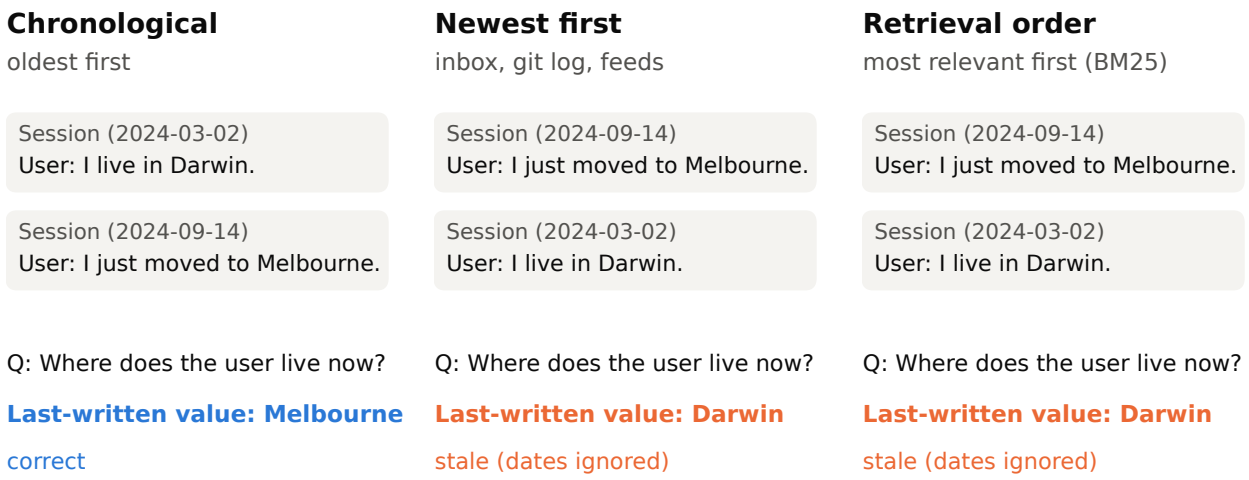


Figure 1: Order versus time. The same two dated sessions shown in three orders (schematic example). A reader that takes the last-written value as current answers correctly only under chronological presentation. Newest-first lists and retrieval-ranked memories are common at deployment; the dates are present in all three cases.

time is usually still *written down*—as a date, a timestamp or a step number—but it disagrees with the order of presentation (Figure 1).

This paper asks a simple question: **when presentation order and explicit time disagree, which one does a language model follow?** Standard benchmarks cannot answer it: their pipelines present updates chronologically, or their questions rarely involve updated states, so that a reader who uses dates and a reader who uses position give the same answer (§6). We therefore hold the content fixed and vary only the order, across controlled chains of thought, deployed formats, third-party long-term memory benchmarks and an official retrieval pipeline, for open models from 1.7B to 32B parameters (two families in the scale study), with and without a reasoning ("thinking") mode.

Our findings are:

1. **Position wins over explicit time.** In shuffled chains of thought, readers answer with the last-presented value in 89.5–99.5% of items once a variable is overwritten eight times; step numbers, "listed from most recent" headers and explicit retractions rarely override this in small models (§4.1). In deployed formats, showing dated LongMemEval sessions newest first costs 24–53 points, and on the held-out MemConflict benchmark [44] Qwen3-4B drops from 75.4% to 36.2% newest first and to 56.2% in BM25 relevance order, an order that a standard retrieval memory produces without any adversarial intervention (§4.2, §4.3).
2. **Neither scale nor reasoning removes it.** From Qwen3-4B to Qwen3-32B the newest-first cost fades where every record carries its own date or the context is short, but stays at 21 points on MemConflict and 23–24 on LongMemEval, where dates sit in session headers far from the facts. In Gemma 3 it *grows* from 12B to 27B (MemConflict −47 to −53; ConvoMem-long −15 to −36). Qwen3's thinking mode removes most of the effect in short conversations but only halves it in long sessions: on MemConflict it leaves −17, −17 and −10 points at 8B, 14B and 32B, where 12% of the 32B answers still state the change in the wrong direction (§5).
3. **A standard retrieval pipeline produces it.** LongMemEval's official retrieval-and-reading code sorts the retrieved sessions by date before the model reads them. Without that sort, knowledge-update accuracy falls by 14–17 points, and the loss falls entirely on the items whose newest evidence BM25 ranks first (Qwen3-4B with thinking: 83.8%→40.5%); sorting restores it,

and thinking does not (§6).

4. **It is specific to conflicting states.** On PersonaMem [22] and LoCoMo [33] temporal questions, which do not require choosing between an older and a newer state, order has no effect at any size (§7).
5. **Two failure points.** In short contexts the failure is one of *selection*: the newer value is linearly decodable at the answer position in ≈98% of items, including those answered with the stale value, and the misleading order is copied into the model's own timeline or thinking, which it then reads by position. In long sessions the failure is one of *binding*: dates in session headers are not attached to facts that appear thousands of tokens later. We find no sign of a semantic "written later means newer" rule; position decides when explicit time is not used (§8).
6. **Using the dates can be trained; deciding when to use them is the bottleneck.** Qwen3-4B models trained to write the dated records of the queried attribute in their thinking answer newest-first MemConflict items as well as chronological ones *when they write the list* (71.7 and 77.9% newest first for the two versions that list almost always; base model with thinking: 50.0%) and recover LongMemEval's newest-ranked-first items (43.2%→73.0%). Versions that write it for every memory question lose 6–17 points on PersonaMem questions about reasons and preference histories; versions that avoid this harm skip the list on the official LongMemEval prompt (6% of thoughts) or answer with one-line guesses (§9).

These results have a practical reading. A pipeline that has timestamps should present retrieved evidence in chronological order: on MemConflict and LongMemEval this recovers the full loss at no cost, and neither prompts nor thinking mode substitute for it. Where no single timestamp per passage exists—extracted facts, summaries, tool outputs from several sources, an agent's own trajectory, a monitor reading another model's reasoning—the model's own reading strategy decides, and that strategy is positional. Benchmarks that present updates chronologically hide this failure.

For every experiment after the initial chain-of-thought study, we wrote a numerical threshold before running it; we report all outcomes, including those that did not meet their thresholds (Appendix A), and every version of the trained procedure that we evaluated on held-out data (Appendix E).

2 Related work

Order and position in context. Models use information at the beginning and end of long contexts better than in the middle [31], and permuting premises hurts reasoning [7]. Studies of shuffled chains of thought report that answers largely survive sentence- or line-level shuffling [8, 46]. Closest to our chain-of-thought results, Liu [30] show that small (1–3B) models read arithmetic CoT by copying whichever number occupies the trailing position, while shuffling breaks BBH tracking and logical deduction and larger models become more selective. Our CoT experiments extend this positional readout to states that are overwritten several times and pit it against explicit time cues; unlike them, we find logical deduction almost unaffected by shuffling (§4.1), consistent with our account that order matters when a state is overwritten. Several studies find the opposite direction of positional preference in other settings: for many in-context updates of a key–value list, retrieval of the latest value degrades and early values intrude [6, 39]; for two short conflicting documents, source framing outweighs position and Gemma shows a primacy bias [15]; and across six types of contextual conflict, including timestamped ones, models attribute more weight to earlier evidence [50]. We discuss how these fit our results in §8.4. Fact order affects temporal reasoning [14], and in-context retrieval shows recency and contiguity biases reminiscent of human episodic memory [4]. When recency and another cue are perfectly correlated in training, which rule a model adopts

is not determined by the data [29]; this is why we make the cues conflict. Dates are not ignored everywhere: as rerankers, models promote passages with injected recent publication dates [13]. None of these studies pits presentation order against explicit time in the reading of updated states.

Knowledge updates and stale state. Entity tracking has been studied with box-moving tasks [24]; Tang et al. [43] find that models do not update world states step by step but aggregate the relevant statements at query time, and fail when the order of operations matters. Concurrently, Guo et al. [17] characterize "stale binding" when a variable is updated in context: the current value remains decodable, attention drifts to older values, and an attention bias at known positions repairs the answer. Their updates are always presented chronologically and the repair requires knowing where the current value is; we study the conflict between order and explicit time, show that the failure is copied into the model's own reasoning, separate selection from binding in long sessions, and test whether the model can learn to use the dates. Outdated retrieved documents can override correct parametric answers, and dates alone barely help [1]; temporal conflicts across document versions mislead models [36]; and masking superseded records in the KV cache shifts preference toward new values at a cost in detail [53]. On the generation side, reflection markers such as "wait" behave as surface cues [51]; we find a similar weakness of retraction markers on the reading side.

Long-term memory. LongMemEval [47] includes knowledge-update questions and sorts retrieved sessions by timestamp in its official pipeline; we quantify what this sort protects against. MemConflict [44] evaluates memory systems under dynamic, static and conditional conflicts but does not vary presentation order. RoutePrism [48] varies the order in which records are *ingested* into a memory and shows that retention policies then keep different evidence; our effect arises with all evidence present. Jiang et al. [23] show that much of the gain of structured memories comes from a readable, time-ordered rendering, in line with our results; in append-only memories, stale chunks can dominate retrieval itself [18]. StateMemBench [12] isolates "state drift", acting on a superseded value that is present in context, and lists "trust the most recent mention" among the lazy reader policies it is designed to trap. Memory systems add explicit state labels [40] or temporal indexes [32], and Supersede [38] trains a bounded-memory agent to keep current values, noting that frontier models resolve explicit updates by a last-mention scan—the heuristic we show fails out of chronological order. Some repairs build the same assumption in: DZ-TDPO [28] adds an attention bias that decays with token distance, assuming that valid updates sit at the trailing edge of the context, which our results suggest would hurt under newest-first or retrieval order.

3 Setup

Formalization. A context contains records $r_1, \dots, r_n$ about one queried quantity. Record r_i carries a value v_i and a time τ_i that is *written in the text* (a date, a timestamp, or a step number), and the records are presented in an order given by a permutation π, where $\pi(i)$ is the position of r_i. The current value is

$$v^\star = v_{i^\star}, \qquad i^\star = \arg\max_i \tau_i. \tag{1}$$

Two idealized readers answer differently:

$$f_{\text{time}}(r, \pi) = v_{\arg\max_i \tau_i}, \qquad f_{\text{pos}}(r, \pi) = v_{\arg\max_i \pi(i)}. \tag{2}$$

They agree whenever π is increasing in τ, which is the case in almost all naturally written text and in standard benchmark pipelines. We therefore compare the same records under several orders:

CHRON (oldest first, π increasing in τ), NEWEST (newest first, π decreasing in τ), shuffled, and RETR, in which records are sorted by a BM25 relevance score $s(q, r_i)$ to the question q, most relevant first:

$$\pi_{\text{retr}}(i) < \pi_{\text{retr}}(j) \iff s(q, r_i) > s(q, r_j). \tag{3}$$

RETR is not chosen by us: it is the order a standard retrieval memory hands to the reader.

Measures. We report accuracy and the *stale rate* (the answer is a superseded value, or states the change in the wrong direction). The *order effect* is the paired difference $\Delta = \text{Acc}(\text{ORDER}) - \text{Acc}(\text{CHRON})$ over the same items, with a 95% paired bootstrap interval (10,000 resamples of items). Because items in the held-out sets share contexts (ConvoMem personas, PersonaMem's 33 shared contexts, MemConflict users, LoCoMo's 10 conversations), we also resampled these clusters; no conclusion changes, and the intervals widen by at most 1.5 points. Free-text answers are graded by Qwen3-14B (4-bit) given the question, the reference answer and, where available, the older and newer evidence, choosing among *up to date*, *stale*, and *other*. On responses that name both values, the judge agrees with a string rule (which value is mentioned first) on 90.8% of Qwen3-4B's and 91.2% of Qwen3-32B's MemConflict answers; on LongMemEval, agreement with string matching is 91.4%. The same judge is used for every model; for the official LongMemEval pipeline we use its official grading prompt with the same judge model.

Models and inference. Chain-of-thought experiments use 14 open models: Qwen3-1.7B/4B/4B-Base/8B/14B [49], OLMo-2-7B (Base, SFT, DPO, Instruct) and OLMo-2-13B (Base, Instruct) [35], Qwen2.5-Math-7B, DeepSeek-R1-Distill-Qwen-7B [11], and Phi-4-mini-instruct [34]; models of 7B parameters and above run in 4-bit with Hugging Face Transformers (HF). The application and mechanism experiments use Qwen3-4B, Qwen3-14B (4-bit) and Phi-4-mini on HF. The scale study (§5) runs Qwen3-4B, 8B, 14B and 32B and Gemma 3 12B and 27B [16] in bf16 on vLLM [26], with identical prompts, decoding rules and judge. vLLM reproduces HF item by item on 97.2% of short items and 94.0% of 37k-token items, with condition means within 2 points; the second figure missed our pre-registered 95% criterion, so we compare models only within one engine and mark every cross-engine comparison (Appendix F). Models answer in non-thinking mode with greedy decoding unless stated. Thinking mode (Qwen3 only) uses Qwen3's recommended sampling (temperature 0.6, top-p 0.95, top-k 20) with a fixed seed per item and a budget of 1,024 thinking tokens; the answer is parsed from the text after the thinking block.

Protocol. For the training study (pre-registered) and for all experiments after the initial chain-of-thought study, we wrote the hypothesis and a numerical threshold before running; in the chain-of-thought study, task categories and conditions were fixed in code before running. Controls differ from treatments in exactly one variable, except where we say otherwise. A phenomenon *replicates* on a set when the order effect is at most -10 points and its interval excludes zero. LongMemEval was used to discover the phenomenon and is reported as *seen*. ConvoMem personas 50–99, ConvoMem-long, ConvoMem-fresh, PersonaMem, MemConflict and LoCoMo are held out and were never used to tune anything; however, we evaluated every version of the trained procedure of §9 on some of them, and MemConflict and PersonaMem informed the *design* of later versions through aggregate failure types (LongMemEval informed the last three). ConvoMem-fresh was built after all design decisions and was used for one version only. We flag design-informed sets wherever we report trained models (Appendix E). Table 1 lists the test sets.

Table 1: Test sets. "Conflict": the question requires choosing between an older and a newer state of the same quantity. Orders: C = chronological, N = newest first, S = shuffled, R = BM25 retrieval order, D/R = official pipeline with and without its date sort. *Held out, but aggregate failure types informed later versions of the trained procedure (Appendix E).

Set	Source	Length (tokens)	Conflict	Orders	Status
Variable tracking in CoT	ours (synthetic)	16–72 lines	yes	C, S, N	–
BBH, model's own CoT (26 tasks)	Suzgun et al. [41]	short	varies	C, S	–
LongMemEval knowledge update (78)	Wu et al. [47]	4.1k–9.3k	yes	C, N	seen
LongMemEval official RAG (500; 78 KU)	Wu et al. [47]	BM25 top-5 sessions	yes (KU)	D/R	seen
E-mail threads, git log, CHANGELOG	ours (synthetic)	0.5k–8.4k	yes	C, N	–
Agent trajectories + memory	ours (synthetic)	short	yes	placement	–
ConvoMem-long (124)	Pakhomov et al. [37]	2.4k (median)	yes	C, N	held out
ConvoMem-fresh (300 + 600)	Pakhomov et al. [37]	2–6 conversations	yes	C, N	held out
MemConflict, dynamic (240)	Tao et al. [44]	6.7k–21.7k	yes	C, N, R	held out*
PersonaMem 32k (589)	Jiang et al. [22]	16k–30k	no	C, N	held out*
LoCoMo temporal (321)	Maharana et al. [33]	13k–25k	no	C, N, R	held out
MemoryAgentBench CR (76)	Hu et al. [21]	6.9k	counterfactual	C, N, R	seen

4 Position is read as time

4.1 Chains of thought

Shuffling hurts only overwritten state. We generate 16-line programs whose CoT assigns the queried variable k times, and ask the reader for its final value after the CoT lines are shuffled (Figure 2a; Qwen3-4B, $n = 200$ per cell). Variables assigned once are read correctly after shuffling ($\geq$99.5%), but accuracy on overwritten variables falls to 79.5/39.5/19.0/14.0% at $k = 1/2/4/8$, close to chance among the k+1 candidates. The reader is not guessing: it picks the value presented last in 93.5–98.5% of items for $k \geq 2$, and in 93–99% of items in a longer variant with up to $k = 64$ overwrites, where the first-presented value is never chosen for $k \geq 2$. At $k = 8$, all 14 models pick the last-presented value in 89.5–99.5% of shuffled items, while variables assigned once stay above 96.8%. Larger models resist at small k (at $k = 2$: Qwen3-14B 73.5%, OLMo-2-13B-Instruct 87.0%) but not at $k = 8$ (97.0% and 95.0%); Qwen3-32B also picks the last-presented value in 97% of items at $k = 8$ (§5). Base, SFT, DPO and RL-tuned OLMo-2-7B checkpoints behave alike (93.5–99.0% at $k = 8$), and a natural-language version ("Box E has 41 marbles") behaves identically. On the model's own reasoning for all usable BBH tasks, shuffling breaks 39% of answers on tasks that we labeled in advance as tracking updated state, and 4% on the others. This is consistent with reports that shuffled CoT preserves answers [8, 46], which concern reasoning whose states are mostly not overwritten: order matters when a state is overwritten. On logical deduction we see a small drop (five objects: 89.6%→85.3%), whereas Liu [30] report a large one; the settings differ (line-level shuffling of the model's own CoT here, 1–3B readers there).

Explicit order cues lose to position. When the CoT lines are listed newest first with a step number on every line (Figure 2b), Qwen3-4B answers correctly in 3–14% of conflict cases, OLMo-2-13B-Instruct in 11.5–26% and OLMo-2-7B-Instruct in 0–1%; of the four models we tested in this condition, only Qwen3-14B uses the numbers (87.5–90.5%), and in the scale study Qwen3-8B and 32B do so as well (62% and 80% at $k = 8$; §5). A header stating that the list is "most recent first" is worse still (Qwen3-4B: 0–1%). A retraction marker attached to the wrong value ("$t = 51$. Hmm, that's wrong.") helps but does not override position: when the retracted value comes last, Qwen3-4B still picks it in 60–80% of items, and the best model (Qwen3-14B) reaches 47.5% accuracy. In real reasoning traces (OpenR1-Math) with two counterfeit conclusions, the earlier one A and a

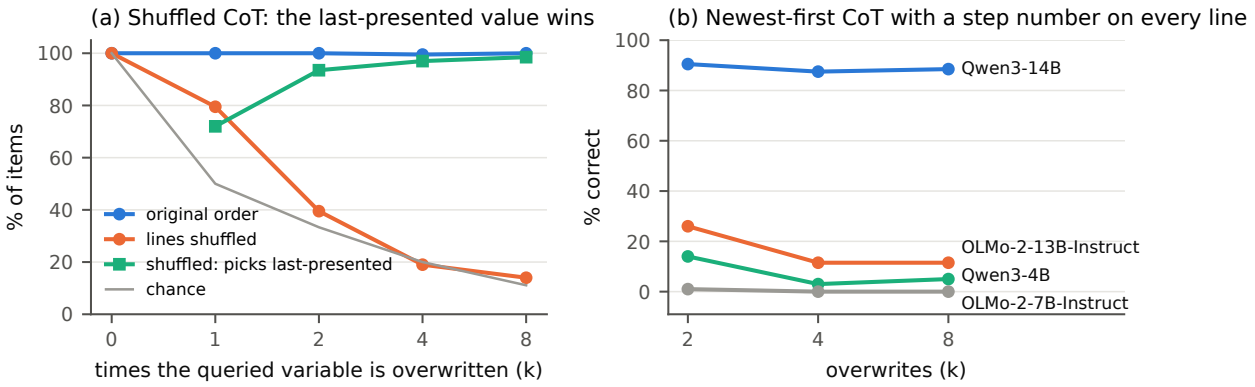


Figure 2: Chain-of-thought readers take the last-presented value. (a) Qwen3-4B on a synthetic 16-line program whose CoT overwrites the queried variable k times ($n = 200$ per point). After shuffling, accuracy falls to chance because the reader picks whichever value is presented last. (b) Newest-first CoT with an explicit step number on every line; accuracy in the conflict case, where position points to a stale value. Among these models only Qwen3-14B uses the step numbers; Figure 4b adds Qwen3-8B and 32B.

later one B followed by "Wait, that's wrong.", a rational reader should choose A; with 10% of the derivation kept, readers still choose the retracted B at least as often (Qwen3-4B 56.5% vs. 19.4%, Qwen3-14B 45.5% vs. 27.5%, OLMo-2-13B-Instruct 41.5% vs. 40.0%). Reading its own unedited reasoning, by contrast, Qwen3-4B is accurate (93–100% on most BBH tasks): the positional default is usually right for text written in chronological order.

4.2 Deployed formats

Chat memory. We take the 78 knowledge-update questions of LongMemEval, each with two dated sessions (4.1k–9.3k tokens) in which the later session updates a fact from the earlier one, and present the sessions oldest or newest first (Figure 3a). Newest first costs Qwen3-4B 25.6 points (76.9→51.3), Qwen3-14B 24.4 and Phi-4-mini 52.6; stale answers rise to 41, 36 and 59%. All intervals exclude zero. Adding "the sessions are listed from the most recent to the oldest" recovers 0–2.6 points; removing the dates makes things worse, so the dates are used, but weakly.

Logs and agent memory. In synthetic e-mail threads, git logs and changelogs in which every entry is dated, newest-first presentation changes accuracy by $-12.1\,[-15.6, -8.6]$ points for Qwen3-4B, -27.4 for OLMo-2-13B-Instruct and -31.4 for Phi-4-mini, and only -3.9 for Qwen3-14B in these short (0.8k-token) contexts. E-mail threads are hardest: Qwen3-4B drops from 88.3% to 58.0%, Phi-4-mini from 86.3% to 15.0%. In agent trajectories, inserting a retrieved, explicitly time-stamped stale memory right before the question lowers Qwen3-4B accuracy by 9.7 points (k = 3 updates: 88.7→69.3), while the same memory at the beginning of the context is harmless (-0.7): what matters is where the stale value sits.

4.3 Held-out benchmarks and retrieval order

ConvoMem-long and ConvoMem-fresh. From ConvoMem's changing-evidence questions [37] we take personas 50–99 (never used in any design decision) and the full 40-message conversations (median 2.4k tokens), date the two conversations in their source order, and present them oldest or newest first. Newest first changes Qwen3-4B's accuracy by $\Delta = -17.7\,[-25.0, -10.5]$ points (95.2→77.4) and Phi-4-mini's by $-62.9\,[-71.8, -54.8]$ (97.6→34.7). Because ConvoMem-long had

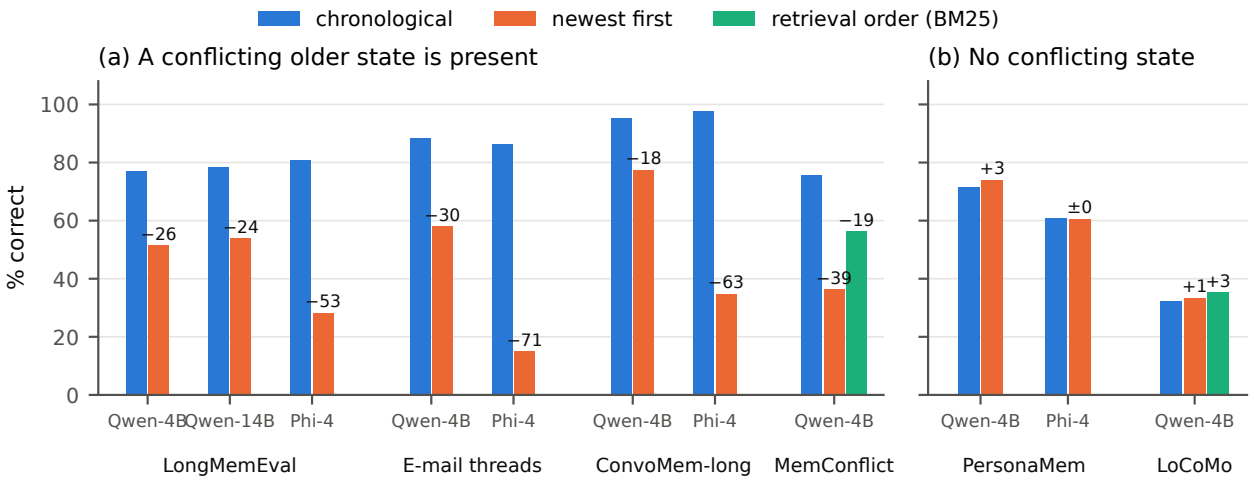


Figure 3: Order effects in deployed formats and third-party benchmarks (Qwen3-4B, Qwen3-14B and Phi-4-mini, HF). Numbers above bars are differences from the chronological bar. (a) When an older, conflicting state is present, newest-first presentation costs 18–71 points, and BM25 retrieval order costs 19 points on MemConflict. LongMemEval and e-mail numbers from §4.2; ConvoMem-long, MemConflict from §4.3. (b) PersonaMem and LoCoMo temporal questions, which do not require resolving such a conflict, show no order effect (§7).

been used to evaluate several trained models, we later built *ConvoMem-fresh* from items of the same personas that no experiment had touched: 300 two-update items and 600 items in which the value changes three to six times, each conversation headed by its date (Appendix C). On vLLM, newest first costs Qwen3-4B $-20.7\,[-25.7, -15.7]$ and $-22.2\,[-25.8, -18.7]$ points, and Qwen3-8B -15.7 and -10.5.

MemConflict. MemConflict simulates users whose state changes over four years of monthly sessions and asks, after each update, how an attribute changed. We use dynamic-conflict questions in which the old value is *not* restated in the update session but appears in an earlier one, so that the answer requires relating two sessions by their dates. Of the 2,946 dynamic-conflict questions, 1,794 are of medium difficulty, 1,208 name exactly one changed attribute, 967 do not restate the old value in the update session, and 850 also mention it in an earlier session; we sample at most 8 per simulated user (240 questions). Each context contains the earlier session, the update session, and the sessions just before the update (2–6 sessions; median 18.5k tokens), each headed by its date. Qwen3-4B answers 75.4% correctly in chronological order, 36.2% newest first ($\Delta = -39.2\,[-46.2, -31.7]$) and 56.2% in BM25 retrieval order ($\Delta = -19.2\,[-25.4, -13.3]$); the share of answers that state the change in the wrong direction rises from 5.0% to 37.9% and 18.3%. For Phi-4-mini, $\Delta = -37.9\,[-45.4, -30.8]$ newest first and $-14.6\,[-20.8, -8.3]$ in retrieval order. Retrieval order places the update session first in 86 of 240 items; accuracy is 48.8% on these and 60.4% on the rest. Since the retrieved sessions are the same in both conditions, re-sorting retrieved memories by their timestamps before reading recovers the full 19 points at no cost.

The selection of items is not what creates the effect. The complementary questions, whose update session itself restates the old value ("moved from A to B"), form a pre-registered control (219 items after the same construction and sampling). The effect is smaller there but still large: newest first gives Qwen3-4B $\Delta = -24.2\,[-31.1, -16.9]$, $15.0\,[3.9, 26.6]$ points less negative than on the main set (user-clustered intervals), and retrieval order costs the same on both sets (difference

Table 2: Order effect by model size and family (bf16, vLLM; same items, prompts and judge). Accuracy change (points) under newest-first (or retrieval) presentation relative to chronological, with 95% paired bootstrap intervals; CoT rows are percentages of items ($k = 8$, $n = 100$). "Thinking": Qwen3's thinking mode; Gemma 3 has none. †Qwen3-4B with thinking on HF (desktop), without an interval (the item-level file is not in our shared record); without thinking on HF, −39.2 and −19.2.

Test	Qwen3-4B	Qwen3-8B	Qwen3-14B	Qwen3-32B	Gemma 3 12B	Gemma 3 27B
MemConflict, newest first	−40.4 [−47.5, −33.3]	−37.1 [−43.8, −30.4]	−35.8 [−42.5, −29.2]	−21.2 [−27.9, −14.6]	−47.1 [−54.6, −40.0]	−52.9 [−60.0, −45.8]
with thinking	−31.7†	−16.7 [−22.5, −10.8]	−16.7 [−22.9, −10.4]	−10.4 [−16.2, −4.6]	–	–
MemConflict, retrieval order	−19.2 [−25.4, −13.3]	−17.9 [−23.8, −12.5]	−15.4 [−21.2, −9.6]	−10.0 [−15.4, −4.6]	−16.7 [−23.3, −10.0]	−18.3 [−25.4, −11.2]
with thinking	−13.4†	−7.9 [−13.8, −2.5]	−14.6 [−20.4, −8.3]	not run	–	–
chronological accuracy	77.1	78.3	84.6	80.0	73.8	75.8
LongMemEval, newest first (seen)	−24.4 [−35.9, −12.8]	−24.4 [−34.6, −14.1]	−23.1 [−33.3, −12.8]	−24.4 [−35.9, −14.1]	−34.6 [−46.2, −21.8]	−42.3 [−55.1, −29.5]
ConvoMem-long, newest first	−17.7 [−25.0, −10.5]	−12.1 [−17.7, −6.5]	−2.4 [−6.5, 1.6]	−6.5 [−11.3, −2.4]	−15.3 [−21.8, −8.9]	−35.5 [−44.4, −27.4]
Dated logs, newest first	−11.3 [−14.9, −7.9]	−13.6 [−17.1, −10.1]	−3.6 [−5.3, −1.8]	−2.9 [−4.7, −1.1]	−6.2 [−9.2, −3.2]	−17.8 [−20.8, −14.9]
Agent, stale memory before question	−9.3 [−12.7, −6.3]	−8.3 [−11.7, −5.3]	−0.7 [−2.0, 0.7]	−0.3 [−1.0, 0.0]	0.0	0.0
PersonaMem, newest first (no conflict)	0.0 [−3.4, 3.4]	−1.7 [−4.6, 1.3]	+3.8 [0.0, 7.6]	−1.7 [−5.5, 2.1]	−0.4 [−4.6, 3.8]	−0.8 [−4.2, 2.5]
CoT shuffled: picks last-presented (%)	98	99	97	97	66	59
CoT newest first + step numbers: correct (%)	4	62	63	80	44	69

+1.8 [−6.8, 9.8]). When the direction of a change is written inside one session, position matters less, but it still matters.

5 Neither scale nor reasoning removes it

We ran the same tests on Qwen3-4B, 8B, 14B and 32B and on Gemma 3 12B and 27B, in bf16 on one engine (Table 2, Figure 4). Before the remaining 32B results were in, we wrote down how a 32B result would change our claims (Appendix A): if the effect replicated on at least half of the sets on which it replicated at 4B, we would claim it for 1.7B–32B and report any attenuation; otherwise we would restrict the claim to models up to 14B.

In Qwen3, the effect persists where dates are far from the facts. On MemConflict the newest-first cost is −40.4, −37.1 and −35.8 points at 4B, 8B and 14B, and −21.2 [−27.9, −14.6] at 32B, where 20.0% of answers still state the change in the wrong direction (3.3% in chronological order). The 32B effect is about half the 4B effect on the same items (+19.2 [11.2, 27.1]); chronological accuracy is similar (80.0 vs. 77.1), so the difference is not a ceiling effect, and the judge is as reliable on 32B's answers as on 4B's (§3). Retrieval order still gives $\Delta = -10.0$ [−15.4, −4.6] at 32B, exactly

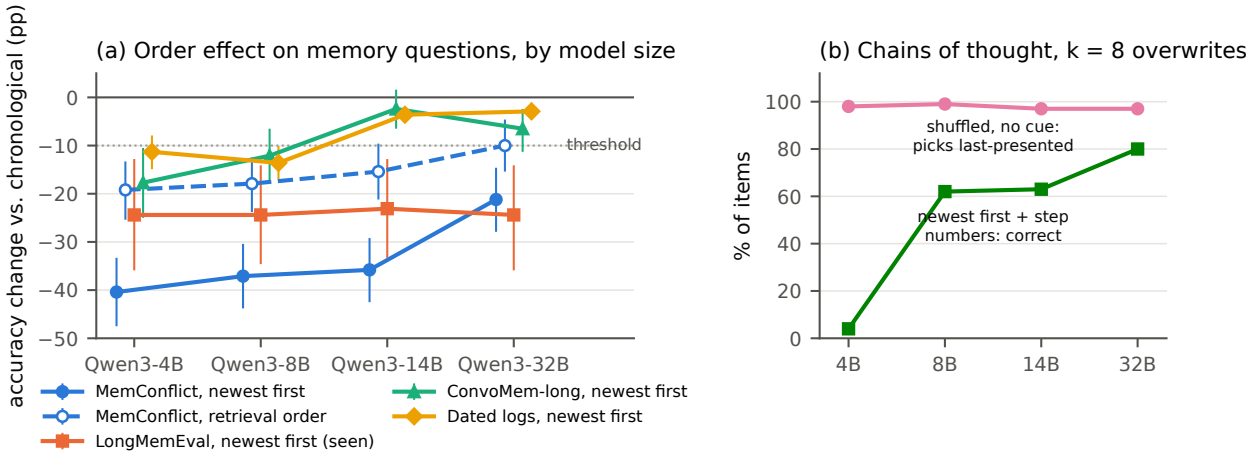


Figure 4: In Qwen3, scale removes the effect only where dates are local (bf16, vLLM, thinking off). (a) Newest-first (and retrieval) minus chronological accuracy with 95% paired bootstrap intervals; the dotted line is our replication threshold. Where every record carries its own date (logs) or the context is short (ConvoMem-long, 2.4k tokens), the effect fades by 14B; where dates sit in session headers thousands of tokens before the facts (MemConflict, LongMemEval), it persists at 32B. (b) Chains of thought: without a time cue, all sizes read the last-presented value; with a step number on every line, larger models increasingly use it.

at our threshold. On LongMemEval the cost does not change with size (-23.1 to -24.4), and at 32B a "most recent first" header still leaves $\Delta = -17.9\,[-28.2, -9.0]$. The 32B effect does not depend on length within MemConflict (terciles with median 14k, 18k and 21k tokens: -21.2, -22.5, -20.0). On dated logs, agent memories and ConvoMem-long, the effect falls below 7 points at 14B and 32B, and in chains of thought larger models use step numbers, but without any cue every size reads the last-presented value (Table 2). We had predicted this pattern before seeing the remaining 32B results (effect below 5 points on dated logs, at least 10 points on long sessions with dates in headers); it held for logs, MemConflict and LongMemEval and failed for ConvoMem-long, whose conversations are short and on which 32B is at ceiling. By the rule written in advance we claim the effect for 1.7B–32B and report that it weakens with scale in short or locally dated contexts.

In Gemma 3, it grows with size. Gemma 3 shows the effect more strongly than Qwen3 of similar size, and the 27B model is worse than the 12B one on every memory and log set: MemConflict -47.1 and -52.9, LongMemEval -34.6 and -42.3, ConvoMem-long -15.3 and -35.5 (99.2% chronological, 63.7% newest first), and dated logs -6.2 and -17.8, mostly because newest-first changelogs collapse (99%→42% at 27B). At the same time, 27B reads short CoT more by time than 12B (step numbers: 69% vs. 44% correct; last-presented value picked in 59% vs. 66% of shuffled items), and neither size is affected on PersonaMem. Scale thus improved the use of local time cues in both families but did not remove the long-context failure in either. We do not extrapolate either trend beyond the sizes tested.

Thinking mode halves the effect in long sessions and removes most of it in short ones. Reasoning models are the obvious remedy, so we ran Qwen3 with thinking on (Figure 5). We pre-registered how to read the result (Appendix A): thinking *removes* the effect if the newest-first effect with thinking is above -5 with an interval including zero, and *weakens* it if it is at least 10 points smaller than without thinking with an interval excluding zero. In ConvoMem-fresh's short conversations, thinking removes most of the effect (Qwen3-4B: $-20.7 \rightarrow -6.7$ and

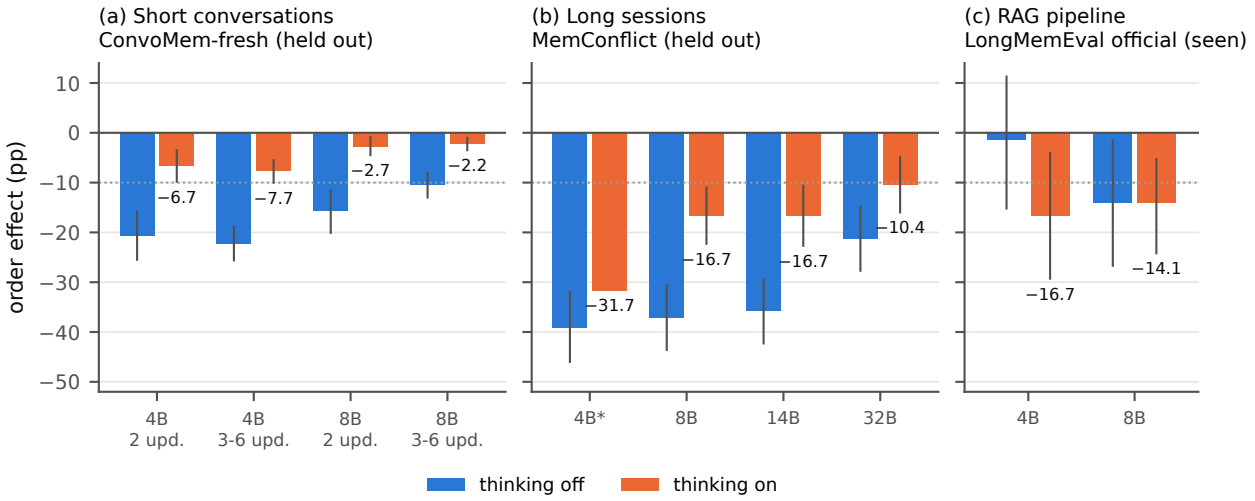


Figure 5: Thinking mode reduces the order effect less the longer and more realistic the context. Order effect (points) with Qwen3's thinking mode off and on; bars show 95% paired bootstrap intervals, the dotted line is our replication threshold, and labels give the thinking-on effect. (a) ConvoMem-fresh, newest first minus chronological, 300 items with two updates and 600 with three to six (vLLM). (b) MemConflict, newest first minus chronological (240 items; 8B–32B on vLLM; *4B on HF, no interval for the thinking-on run). (c) LongMemEval's official retrieval-and-reading pipeline, relevance order minus date order, 78 knowledge-update questions (vLLM; LongMemEval is seen).

$-22.2 \to -7.7$; Qwen3-8B: $-15.7 \to -2.7$ and $-10.5 \to -2.2$), although small effects remain (all four intervals exclude zero); this comparison was planned as a report without a threshold. On MemConflict, thinking weakens but does not remove it at any size: the effect is $-16.7\,[-22.5, -10.8]$ at 8B, $-16.7\,[-22.9, -10.4]$ at 14B and $-10.4\,[-16.2, -4.6]$ at 32B, against -37.1, -35.8 and -21.2 without thinking (reductions of $+20.4$, $+19.2$ and $+10.8$ points, all intervals excluding zero). By our replication rule the effect still replicates at all three sizes, at 32B by 0.4 points; 12.1% of the 32B answers with thinking state the change in the wrong direction newest first, against 3.8% chronologically. At 4B thinking helps little ($-39.2 \to -31.7$, HF). In retrieval order, thinking weakens the effect at 8B ($-17.9 \to -7.9$) but not at 14B ($-15.4 \to -14.6$). Unfinished thoughts are rare (0.3–2.8%). In the official retrieval pipeline of the next section, thinking does not help at all.

6 A standard retrieval pipeline

Our MemConflict contexts contain only the sessions relevant to the question. To test a complete pipeline that we did not build, we ran LongMemEval's official retrieval-and-reading setup [47]: BM25 retrieves the top five sessions of each of the 500 questions' haystacks, the official reading prompt presents them, and the official grading prompt (with our judge model) scores the answer. The official code sorts the retrieved sessions by date before reading; we compare this with the same sessions in relevance order, which is what the reader sees whenever the sort is skipped or no timestamps are available. LongMemEval is seen data; we report it as a test of the pipeline, not of any repair.

Relevance order costs 14–17 points on knowledge updates. Without the date sort, knowledge-update accuracy falls by 16.7 points for Qwen3-4B with thinking and by 14.1 points for Qwen3-8B with and without thinking (Table 3); accuracy over all 500 questions changes little

Table 3: LongMemEval's official retrieval pipeline, with and without its date sort (Qwen3, vLLM; accuracy %, date order / relevance order; Δ = relevance minus date with 95% paired bootstrap interval). KU: the 78 knowledge-update questions. Lower block: KU items with at least two evidence sessions in the top five, split by whether BM25 ranks the newest evidence first (from a local reconstruction of the official retrieval, aligned with the cloud outputs).

Reader	All 500 questions	KU (78)	Δ KU
Qwen3-4B, thinking off	57.8 / 58.8	65.4 / 64.1	$-1.3\,[-15.4, 11.5]$
Qwen3-4B, thinking on	66.8 / 66.2	79.5 / 62.8	$-16.7\,[-29.5, -3.8]$
Qwen3-8B, thinking off	64.4 / 60.4	78.2 / 64.1	$-14.1\,[-26.9, -1.3]$
Qwen3-8B, thinking on	67.0 / 63.8	82.1 / 67.9	$-14.1\,[-24.4, -5.1]$
KU items, date order / relevance order			
newest evidence ranked first (37): 4B thinking	83.8 / 40.5		
newest evidence ranked first (37): 8B thinking	86.5 / 51.4		
newest evidence ranked later (30): 4B thinking	80.0 / 86.7		

(+1.0 to −4.0). Qwen3-4B without thinking is the exception ($-1.3\,[-15.4, 11.5]$): its accuracy is low in both orders, and thinking raises it only when the sessions are sorted (65.4→79.5% by date, 64.1→62.8% by relevance).

The loss is exactly where the newest evidence is ranked first. Retrieval is not the problem: 67 of the 78 knowledge-update questions have at least two evidence sessions among the top five, and the same sessions are present in both orders. Among these, the 37 items whose newest evidence BM25 ranks *before* the older evidence account for the entire loss (Qwen3-4B with thinking: 83.8%→40.5%; Qwen3-8B: 86.5%→51.4%), while the 30 items whose newest evidence comes later do not lose (80.0%→86.7%). On Qwen3-8B the date sort does not hurt other question types (all questions 67.0 vs. 63.8%; temporal reasoning 57.9% in both orders). The pipeline thus reproduces our controlled result: the reader takes the later-presented session as the newer one, and the date sort in the official code is what protects it.

Other official pipelines. LoCoMo's official evaluation code presents sessions newest first to both open and API models; there this costs at most 1.5 F1 points relative to chronological order (Qwen3-4B without thinking $-1.2\,[-2.3, -0.2]$, with thinking $-1.5\,[-3.2, 0.2]$), because its questions rarely concern updated states (§7). ConvoMem's official setup keeps the evidence in source order and gives no dates. The standard pipelines we examined therefore avoid, rather than test, the conflict between position and time.

7 Where order does not matter

PersonaMem and LoCoMo. On the 238 PersonaMem questions about how a preference evolved and why it changed (16k–30k tokens), presenting sessions newest first has no effect (Qwen3-4B $+2.5\,[-0.8, 5.9]$; Phi-4-mini $-0.4\,[-3.8, 2.9]$); the same holds for the five other question types (largest drop −5.9, $n = 17$), for every model size and family in Table 2 (−1.7 to +3.8), and for the 321 LoCoMo temporal questions (Qwen3-4B $+1.2\,[-3.7, 5.9]$ newest first, $+3.1\,[-1.6, 7.8]$ in retrieval order; Qwen3-32B $-2.8\,[-7.2, 1.6]$ and $-2.5\,[-7.5, 2.5]$). Diagnostics explain why (Appendix B). PersonaMem's update questions can be answered without the history: matching words between the user's final message and the options alone selects the correct option for 80.8% of the "why did the

preference change" questions (Qwen3-4B with the full history: 83.8%), and Phi-4-mini answers "how did the preference evolve" questions equally well with and without the history (51.8% both). One cell does not fit cleanly and is reported as unresolved (Appendix B). The remaining question types do depend on the history, but none requires choosing between an older and a newer state; neither do LoCoMo's temporal questions, which ask when an event happened (we have not validated the judge on LoCoMo's free-form date answers as we did on the other sets). These null results delimit the phenomenon rather than contradict it: order matters when a conflicting older state is present and time is expressed only by position or by dates far from the facts.

A confound. On MemoryAgentBench's conflict-resolution facts [21], where the newer fact is counterfactual and the older one is world knowledge, Qwen3-4B picks the older fact in 83% of chronological items and Qwen3-32B in 82%: parametric knowledge dominates and the set cannot test order, in its 6k- or its 37k-token version (it was also used in our pre-registered training evaluation, so we treat it as seen). The BEAM benchmark [45] could not be used either: in 168 of its 180 knowledge-update questions the old and the new information share one dated session, so no explicit time separates them.

8 Why: two failure points

8.1 Misleading order is copied into the model's own reasoning

A natural defense is to let the model reason first. In long contexts this helps less than one would hope, because the reasoning inherits the presentation order.

Prefilled timelines. We write a correct, dated timeline of the two pieces of evidence into the model's response and let it only produce the final answer (Qwen3-4B, Qwen3-14B, $n = 70$). When the timeline lists the evidence newest to oldest and the input is newest first, both models still answer the stale value in about 40% of items (40.0% and 38.6%; Phi-4-mini: 57.1%), although every date in the list is correct; with an oldest-to-newest list the rate falls to 27.1% and 10.0%. Left to their own devices, models mostly list evidence in presentation order.

Swapping thoughts. In thinking mode on LongMemEval, Qwen3-4B answers 80.8% correctly from chronological input and 55.1% from newest-first input (Figure 6a). Its thinking restates the evidence in presentation order: among traces that mention both session dates, the newer date comes first in 82.1% of newest-first runs and 3.6% of chronological runs. Swapping the thinking between runs swaps the accuracy: newest-first input with the thinking from the chronological run gives 80.8%; chronological input with the thinking from the newest-first run gives 57.7%; removing the input entirely changes little (75.6% and 52.6%). The answer follows the thinking, and the thinking has absorbed the misleading order. In e-mail threads, thinking fixes the order in short contexts (10 messages: 93.3% correct newest first) but not in longer ones (200 messages: 53.3%), where the last value mentioned in the thinking is the newest one in only 51.7% of items. This matches the pattern of Figure 5: thinking repairs short contexts and leaves long ones.

8.2 Short contexts: known but not used

We train linear probes on the residual stream at the last token before the answer to predict whether the newer value was presented first (layers selected on training items only; 60/40 split by question). On shuffled, step-numbered CoT the probe reaches 98.8% (layer 21), and it decodes the correct

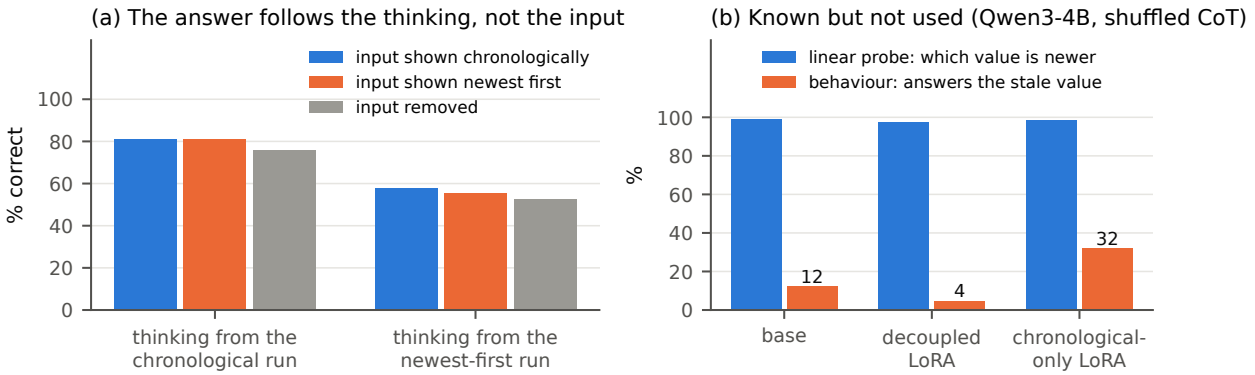


Figure 6: Mechanism. (a) Qwen3-4B in thinking mode on LongMemEval ($n = 78$): accuracy depends on which run the thinking comes from, not on the order of the input it is paired with, or on whether the input is shown at all. (b) On shuffled, step-numbered CoT, a linear probe at the answer position decodes which value is newer in ≈98% of items for the base model and for two LoRA variants trained on decoupled and on chronological-only data (Appendix D); behavior differs widely. Training changes use, not knowledge.

order on all 13 items that Qwen3-4B answers with the stale value. On ConvoMem short dialogues it reaches 97.9%; Qwen3-14B answers MemoryAgentBench fact pairs stale in 50% of items while its probe reaches 94%. The representation is format-specific: probes transfer across formats at near chance (49–58%). In short contexts, the model knows which value is newer and fails to select it. A pre-registered training study confirms that this is a failure of use (Appendix D): LoRA training on synthetic records whose order is decoupled from their dates, in formats that appear in no test, transfers to newest-first logs (70.3%→100%) and to step-numbered newest-first CoT (6.0%→74.7%), whereas the identical data in chronological order does not (89.3% and 2.0%). All three models decode which value is newer equally well (≈98%), but their stale rates are 12.2% (base), 4.5% (decoupled) and 31.8% (chronological only). This training does not reach long sessions (LongMemEval newest first: 48.7% vs. 51.3% for the base model).

8.3 Long sessions: dates are not bound to facts

In long LongMemEval sessions, the same probe is weak (best layer 68.6% for Qwen3-4B and 72% for Qwen3-14B in five-fold cross-validation), suggesting that "which update is newer" is not available at the answer position. Models that we trained to write a dated list before answering (Appendix E, E13) name both session dates in only 19.2% of newest-first items; when they do, they answer correctly in 86.7% of items, otherwise in 46.0%. The dates sit in session headers thousands of tokens before the facts they govern, and the model does not attach them. The information is nevertheless readable: forcing attention onto the newer statement at the answer position, an oracle intervention that requires knowing where the answer is and that we use only as a diagnostic [following 17], raises newest-first LongMemEval accuracy from 52.9% to 72.9% (e-mail threads: 60.7% → 100%). The scale and thinking results fit this account: larger models and thinking stop falling back on position when each record carries its own date or the context is short, but not when the date sits in a header far from the fact (§5). This is a post-hoc reading of a correlational pattern, not a test.

8.4 Position, not a particular direction

Every test above asks for the *current* value, so a reader that maps later text to later time and a reader that simply favors whatever is closest to the answer make the same prediction. Questions

about the *earliest* value separate them: on newest-first input, a semantic "written later means newer" rule picks the last-presented record for the current value and the first-presented one for the earliest value, while mechanical recency picks the last-presented record for both. In a pre-registered test (dated calendar and audit tables with 2–4 records, 150 items per cell, two formats never used elsewhere), no model meets the semantic criterion (current: last minus first ≥ 20 points; earliest: ≤ -20): Phi-4-mini favors the last-presented record for both questions (+36.7 and +18.0, just short of the mechanical criterion), Qwen3-8B does so for the earliest value only (+2.7 and +28.0), Qwen3-4B and Gemma 3 12B and 27B are mixed, and Qwen3-14B and 32B mostly answer by date in these short tables (current −53.3 and −54.0, earliest +53.3 and +56.7), although newest-first order still costs 32B 20.0 points on the current value. Qwen3-4B, in these short tables, even favors the *first*-presented record under reversal. This is how we read the opposite findings in the literature. Primacy-like preferences appear for two short conflicting documents [15, 50] and for long lists of in-context key–value updates [6, 39]; recency appears in our overwritten CoT states and long multi-session memories. The direction of the positional preference depends on the setting; what our results show consistently is that, when the model does not use explicit time, *position* decides. We therefore claim that position overrides explicit time, not that models interpret later text as later time.

9 Can a model learn to use the dates?

The binding account suggests a direct test: teach the model to write down the dated records of the queried attribute before answering, and see whether order still matters. We report this as an analysis of when such a procedure works, not as a finished repair; every version below was specified, with thresholds, before it was trained and tested (one decoding change on Qwen3-8B, noted below, was not), and Appendix E lists all of them.

The procedure and its versions. We generate long synthetic documents organized in blocks, each opened by a single date line and followed by unrelated prose in which statements about attributes are buried (diary, meeting minutes, lab notebook, status report, field log; median 2.8k tokens), with block order decoupled from time (chronological, reversed or shuffled). Questions ask for the current value, the earliest value, the value as of a date, the most recent change, the history or the reason of a change, or concern attributes recorded once or never. In half of the samples, the target thinking lists the queried attribute's records oldest to newest with their dates, followed by one sentence of inference; the other half has no thinking. We train Qwen3-4B with LoRA (rank 16, $\alpha = 32$, one epoch) on these data, mixed with 1,000 samples with local dates. The versions differ in what else they contain (Table 4): E18 is this recipe; E18.1 adds reason questions whose thinking quotes the relevant record without a list, sentence-level answer options, and 474 general instruction responses written by the base model itself [11.9%; following 2, 27]; E19 adds 1,000 synthetic multi-session chats in three formats, presented chronologically, reversed, shuffled or relevant-first; E19.1 makes every memory-question thinking start with the list; E19.2 instead replaces the short thinking of reason and absent-attribute questions with the base model's own thinking on the same prompts. With thinking off, every version that we checked keeps general ability within our pre-registered limits (MMLU [19], ARC-Challenge [9], HellaSwag [52], IFEval [54], LongBench [3] and GSM8K [10], run with lm-evaluation-harness [5]; E18, E18.1, E19 and E19.1; Appendix E).

Table 4: When the dated list is written (Qwen3-4B with thinking on; accuracy %, chronological / newest first / retrieval). "List": share of thoughts that list dated records; "one-line": share of thoughts under 40 words without a list. LongMemEval RAG: the 37 official-pipeline items whose newest evidence is ranked first, relevance order. Base: MemConflict and LongMemEval with thinking on, PersonaMem without thinking. *Design-informed held-out set; LongMemEval is seen and informed E19–E19.2. Desktop HF, except E18.1 seed 0 on MemConflict and PersonaMem (cloud HF). E19.2's PersonaMem evaluation was paused after 68 of 238 items.

	MemConflict*		LongMemEval RAG (37)		PersonaMem*
Model	C / N / R	list / one-line	accuracy	list	C / N
Base	81.7 / 50.0 / 68.3	0% / 1%	43.2	–	71.4 / 73.9
E18	72.9 / 71.7 / 70.0	96% / 4%	73.0	95%	63.9 / 56.7
E18.1, seed 0	80.4 / 77.5 / 77.9	–	45.9	6%	71.4 / 71.0
E18.1, seed 1	70.4 / 62.1 / 70.8	4% / 64%	–	–	–
E19	70.0 / 70.4 / 70.8	31% / 50%	67.6	–	73.1 / 71.0
E19.1	78.3 / 77.9 / 78.8	99% / 0%	73.0	78%	65.5 / 65.5
E19.2	79.2 / 74.2 / 78.3	3% / 4%	–	–	paused

When the list is written, order stops mattering. Table 4 and Figure 7 summarize the versions. The base model with thinking answers newest-first MemConflict items far worse than chronological ones (50.0% vs. 81.7%). E18, which writes the list in 96% of its thoughts, answers them as well as chronological ones (71.7% vs. 72.9%), and so does E19.1 (77.9% vs. 78.3%; list in 99%). On LongMemEval's 37 newest-ranked-first items, E18 and E19.1 reach 73.0% against 43.2% for the base model (seen data). Within a model, the list predicts success: E19 lists in 31% of its MemConflict thoughts and is correct in 80% of these, against 62% for its one-line thoughts; E18's remaining errors on the 37 items are errors inside the list (the newest date is listed in 80% of them, but correctly ordered in only 60%). The long-session failure is therefore not a missing capacity: the dates can be bound when the model writes them down.

Deciding when to write the list is the bottleneck. The versions differ mainly in *when* they write the list. E18 writes it for every question, including PersonaMem questions about the reasons for an update or the evolution of a preference, where "list the records and take the newest" answers the wrong question; it loses 7.5 and 17.2 points there (98% of its PersonaMem thoughts open with the list). E18.1 removes this harm (71.4/71.0%) but, on the official LongMemEval prompt—JSON chat histories and "answer step by step"—writes the list in only 6% of its thoughts and scores like the base model (45.9% vs. 43.2%), while on our own MemConflict and ConvoMem prompts it lists in 20–28%. E19 adds chat data and recovers most of the LongMemEval items (67.6%) without PersonaMem harm, but half of its MemConflict thoughts become one-line guesses ("No change.", "No record mentions . . . "), a shortcut that we traced to the one-sentence training thoughts of reason and absent-attribute questions. E19.1 removes the shortcut by starting every memory thought with the list (one-line thoughts 50%→0%; MemConflict +7.9 [4.2, 11.7] points over E19, averaged over orders) and reaches 73.0% on the LongMemEval items, but lists in 45% of its PersonaMem thoughts and loses 5.9 and 8.4 points there; on reason questions, its listed thoughts are correct in 65% of items against 81% for its long, unlisted thoughts (the model chooses where to list, so this comparison is not causal). E19.2, which takes the reason and absent-attribute thoughts from the base model instead, keeps MemConflict at 79.2/74.2/78.3% with one-line thoughts in 4% and lists in only 3% of

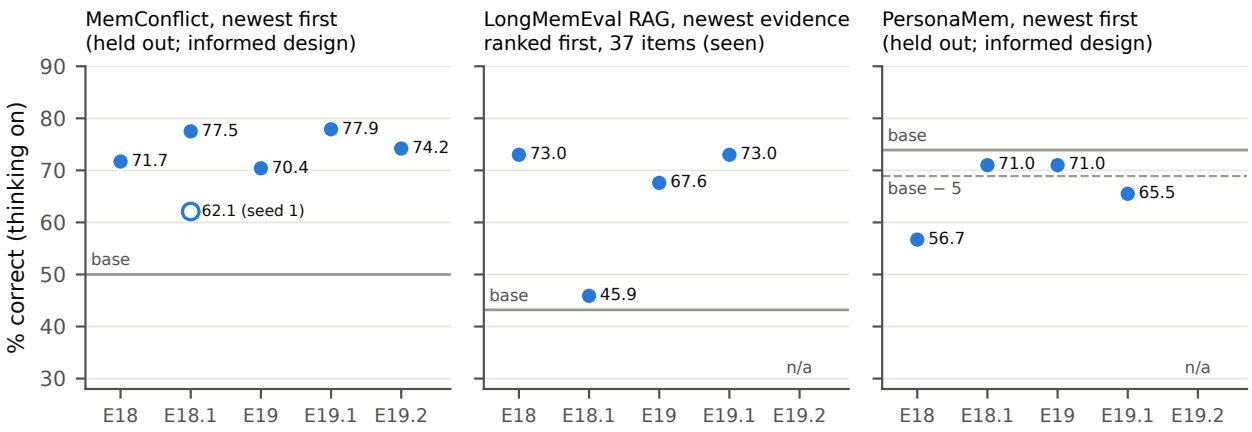


Figure 7: The trade-off across versions of the trained procedure (Qwen3-4B, thinking on; numbers from Table 4). Versions that write the dated list for memory questions (E18, E19.1) recover newest-first MemConflict and LongMemEval's newest-ranked-first items but lose on PersonaMem, whose questions ask about reasons and preference histories; E18.1 keeps PersonaMem but does not write the list on the official LongMemEval prompt. The open circle is a second seed of E18.1, trained and tested on another machine. The PersonaMem base line is the base model without thinking (the thinking-on base run is incomplete), with our pre-registered 5-point limit dashed.

thoughts; its PersonaMem and LongMemEval evaluations are not finished. Getting the list written when, and only when, the question concerns a changed state remains unsolved in our hands.

How much training adds beyond thinking, and how stable it is. On ConvoMem-fresh, the only set that no design decision touched, E18.1 with thinking improves on the base model with thinking by $+5.3\,[3.0, 7.8]$ newest-first points for three to six updates and by $+0.7\,[-2.7, 4.0]$ for two updates; most of the repair on short conversations comes from thinking itself. On MemConflict, the trained versions improve newest-first accuracy over the base model with thinking by 12–28 points, but MemConflict informed their design, and two seeds of E18.1 trained on different machines differ by up to 15 points (77.5% vs. 62.1% newest first); seed and machine are confounded. The same recipe on Qwen3-8B missed its PersonaMem threshold (67.6% vs. 75.6% newest first) and met it only after we added a decoding penalty chosen after seeing that result (Appendix E).

10 Discussion

What the model does. Our results suggest a reading strategy that is reasonable for the data models are trained on: when several values compete and explicit time is not used, the position of a value decides, and in long, overwritten states this means taking the one written last, because text is almost always written in the order things happen. Explicit time is used when it is local and salient—increasingly so with scale and with thinking (step numbers, dated log entries, short conversations)—but in long contexts dates are not bound to the facts and position decides, in Qwen3 from 4B to 32B and in Gemma 3 more so at 27B than at 12B. Reasoning does not escape the default by itself: the thinking restates the evidence in the presented order and is then read by position. A written list of dated records does escape it, which is why we locate the remaining difficulty in recognizing when a question concerns a changed state.

Practical implications. Pipelines should present retrieved evidence in chronological order whenever timestamps are available; on MemConflict this recovers 10–19 points at every size we tested, and in LongMemEval's official pipeline the date sort is what prevents a 14–17-point loss on knowledge updates. Statements about the order, instructions to reason and thinking mode are not substitutes. Sorting is not always possible: memories built from extracted facts or summaries carry no single timestamp per passage; tool outputs from several sources, agent trajectories interleaved with retrieved memories and transcripts read by monitors have no common clock; and in append-only memories, stale entries can dominate what is retrieved [18]. In these cases the model's own reading strategy decides. Evaluation should therefore include non-chronological orders: the official pipelines we examined sort by date, keep source order without dates, or ask questions that rarely involve updates, so that a positional reader and a temporal reader score alike.

Limitations.

- **Scale and families.** Our largest model has 32B parameters; we tested two open families and no API models, and we do not extrapolate the opposite trends of Qwen3 and Gemma 3 beyond the sizes tested.
- **Judges.** Free-text answers are graded by Qwen3-14B, of the same family as most readers, with about 91% agreement with string rules where both apply; we have not validated the judge against human labels or against a judge from another family.
- **Engines and machines.** The scale and thinking studies run on vLLM, most application experiments and all trained models on HF; item-level agreement between the engines is 94–97%. The 14B and 32B thinking runs used concurrent requests, which changes numerics slightly. Qwen3-4B with thinking on MemConflict ran on HF on a different machine from the other sizes (Appendix F).
- **Held-out reuse and seen data.** We evaluated eight versions of the trained procedure (thirteen trained models, counting controls, seeds and the 8B run) on held-out sets; MemConflict and PersonaMem informed their design, LongMemEval informed the last three, and only ConvoMem-fresh has informed no design decision. The LongMemEval pipeline results come from seen data.
- **Sample sizes.** LongMemEval's knowledge-update set has 78 items (37 with the newest evidence ranked first), ConvoMem-long 124; several key effects (32B with thinking, 32B in retrieval order) sit at our threshold.
- **Constructed orders.** Newest-first presentation is constructed by us, although it mirrors common interfaces; retrieval order is not, and it costs 10–19 points on MemConflict and 14–17 on LongMemEval's knowledge updates.
- **The trained procedure** is evaluated with one seed per version except E18.1, whose two seeds differ by up to 15 points; E19.2's evaluation is incomplete.
- **Parsing.** For PersonaMem we corrected a parser after seeing that it failed on answers ending in "The best answer is (d)."; the rule uses only the format, and no prediction of the base models changed.

11 Conclusion

When presentation order and explicit time disagree, language models follow the order. The default is cheap and usually right. Scale and thinking teach models to use dates that sit next to the facts or appear in short contexts, but not dates in session headers thousands of tokens away, so the default survives in exactly the long, retrieved contexts where order is least reliable. A standard retrieval

pipeline loses 14–17 points on knowledge updates when it does not sort by date. The failure has two parts, selecting the newer value and binding facts to their dates; a model can learn to bind them by writing the dated records down, but learning when to do so is the open problem.

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A Pre-registered thresholds and outcomes

Thresholds were written in `EXPLORE_PLAN.md`, `PREREG_TRAINING.md` or `CLOUD_NOTEBOOK.md` before each run. Experiments are Qwen3-4B, seed 0, unless noted. "Met" and "not met" refer to the written threshold.

Experiment	Threshold (written before running)	Outcome
Selection LoRA, primary endpoint	LongMemEval newest first: decoupled $\geq$ control +10, CI > 0	**Not met** (+3.8)
Selection LoRA, transfer	two further test families, CI > 0	Met (logs, CoT, MAB)
Selection LoRA, side effects	chronological conditions and general ability drop $\leq$2	Met at 4B; TempReason -7.0 vs. control; **not met at 1.7B** (GSM8K -58.8)
E1 prefilled timelines	newest-first input raises stale rate $\geq$15 for both models	Met (+28.6 each)
E1c thinking (first reading)	thinking conclusions $\geq$15 more stale	Not met; **corrected** by swap experiment (grader read only the last 800 characters)
E1c swap	answer follows thinking	Met
E1d sorted timeline instruction	stale rate $\geq$15 below the plain timeline prompt and gap to chronological $\leq$5	Not met (gap +21.8)
E2 list reading, Qwen3-4B	decoupled $\geq$ control +10	Met (+25.7; own timeline +19.2)
E2 list reading, Phi-4-mini	same	Met (+27.1)
E3 zeroing recency heads	drop $\geq 2\times$ random and $\geq$10	Not met; confounded (reading also breaks)
E4 primacy at length	first-value rate +20 up to 19k tokens	Not met (0%)
E5 CoT-only LoRA	$\geq$10 on two application tests	Met; LongMemEval reversed (-20.5)
E7 probes, short contexts	$\geq$80% on stale-answered items	Met (100%, $n = 13$)
E7 probes, long sessions	$\geq$80% or $\leq$60%	Inconclusive (69%)
E10 attention routing (oracle)	+15 on a long-dialogue set	Met (LongMemEval +20)
E11 answer from notes only	$\geq$ thinking +15	Not met
New long-dialogue sets	newest first $\leq$ chron. -10, CI < 0	ConvoMem-long met; PersonaMem not
External sets	same	MemConflict met (4B, Phi); LoCoMo not (predicted); MAB-CR untestable
E15 earliest-value questions	semantic: current $\geq +20$ and earliest ≤ -20; mechanical: both $\geq +20$	Mixed for all seven models (Qwen3 4B–32B, Phi-4-mini, Gemma 3 12B and 27B)
E16 restated-value control	restated $-$ main $\geq +15$, CI > 0	Met, at the threshold (+15.0)
Scale: 32B replication	same rule per set	Met: MemConflict (both orders; retrieval at -10.0), LongMemEval, CoT; not: ConvoMem-long (ceiling), logs
Scale: claim at 32B	replicate on $\geq$ half of the 4B-replicated sets	Met $\rightarrow$ claim for 1.7B–32B, weaker with scale
Scale: prediction	32B logs < 5; long header-dated sessions ≥ 10	Met for logs, MemConflict, LongMemEval; not for ConvoMem-long
Thinking mode (MemConflict newest first; 8B, 14B, 32B)	replicates if ≤ -10, CI < 0; removes if > -5, CI $\ni 0$; weakens if thinking $-$ off $\geq +10$, CI > 0	Replicates and weakens at all three sizes (-16.7, -16.7, -10.4; +20.4, +19.2, +10.8)
Thinking mode, retrieval order (8B, 14B)	same	8B: not replicated, weakens (-7.9); 14B: replicates, no clear effect (-14.6)

Experiment	Threshold (written before running)	Outcome
ConvoMem-fresh, E18.1 (thinking)	$\lvert N-C\rvert \leq 5$; $N \geq$ base (off) +10, CI > 0; $C \geq$ base -5	Three to six updates: all met; two updates: first missed (-5.7), others met
LongMemEval official, 4B	relevance $-$ date on KU (base, off) ≤ -10	Not met (-1.3); with thinking -16.7 (report only)
LongMemEval official, 8B	same	Met (-14.1; with thinking -14.1)
LongMemEval official, repair	E18.1 (8B: with penalty) KU in relevance order $>$ base, CI > 0; own relevance $-$ date ≥ -5	**Not met** at 4B (+2.6; own -16.7) and 8B (+6.4; own -14.1)
LongMemEval official, no harm	date order: E18.1 $\geq$ base -5	Met (4B +0.6, 8B -0.8)
LoCoMo official, E18.1	thinking-on F1 gain, CI > 0; lower CI bound ≥ -2	Not met ($-1.2\,[-3.1, 0.7]$)
Fix diagnostic: list needed?	answer-only variant $\geq$10 below list on a long set	Met (LongMemEval -16.7; ConvoMem-long -0.8)
E18	format $\geq$90; ConvoMem-long; PersonaMem $\geq$ base -5; MemConflict; general ability	Met except PersonaMem with thinking on (-7.5/-17.2) and one format cell with thinking off (89.7)
E18.1	PersonaMem $\geq$ base -5; keep ConvoMem-long and MemConflict; format $\geq$90; general ability	Applied tests and general ability (thinking off) met; **format not met** (as-of-date content)
E18.1, Qwen3-8B	same, against base 8B	**PersonaMem not met** (-5.9/-8.0); met with presence penalty 1.5 chosen after the result (flagged); general ability met
E18 on LongMemEval RAG (37 items)	list $\geq$50% and $\geq$ E18.1 +15	Met (95%; 73.0 vs. 45.9)
E19	LongMemEval RAG $\geq$ 73.0, list $\geq$50%; PersonaMem; ConvoMem-long; MemConflict C $\geq$70.4, N $\geq$72.5; general ability	LongMemEval not met (67.6); MemConflict not met (70.0/70.4); others met
E19.1	one-line thoughts $\leq$15% and $\geq$ E19 +3, CI > 0; E19 thresholds	Shortcut and LongMemEval met (0%; +7.9; 73.0); **PersonaMem not met** (65.5/65.5); general ability met
E19.2	PersonaMem; keep E19.1's MemConflict gain; E19 thresholds	MemConflict met (4%; +6.8 over E19); PersonaMem (68 of 238 items), LongMemEval and general ability not finished (paused)

B PersonaMem diagnostics

Table 6 reports accuracy without the history (*bare*: final user message and options; *no history*: persona, message and options) and with the full history in both orders. By the threshold written in advance, a question type "does not depend on the history" if chronological accuracy exceeds no-history accuracy by less than 10 points or the interval includes zero. "How did the preference evolve" questions do not depend on the history for either model. For "why did it change" questions, Qwen3-4B's no-history condition is $12.1\,[4.0, 20.2]$ points below the full history, which formally

counts as dependence; the bare condition (80.8%), however, is within 3 points of the full history, and the persona text lowers accuracy. We note that this reading was made after seeing the results.

Table 6: PersonaMem 32k accuracy (%) by question type. Bare / no history / chronological / newest first.

Question type	n	Qwen3-4B	Phi-4-mini
Track preference evolution	139	56.8 / 58.3 / 62.6 / 67.6	51.8 / 51.8 / 51.8 / 51.1
Reasons behind updates	99	80.8 / 71.7 / 83.8 / 82.8	68.7 / 70.7 / 73.7 / 73.7
Recall user-shared facts	129	40.3 / 48.1 / 59.7 / 56.6	31.0 / 41.1 / 46.5 / 48.1
Recall facts mentioned	17	11.8 / 17.6 / 58.8 / 52.9	41.2 / 41.2 / 70.6 / 70.6
Generalize to new scenarios	57	21.1 / 38.6 / 63.2 / 57.9	17.5 / 28.1 / 43.9 / 38.6
Preference-aligned recommendations	55	40.0 / 23.6 / 58.2 / 61.8	34.5 / 34.5 / 47.3 / 47.3
Suggest new ideas	93	12.9 / 10.8 / 10.8 / 9.7	15.1 / 12.9 / 22.6 / 17.2

C Test-set construction

ConvoMem-long. ConvoMem changing-evidence items with two evidence conversations, personas 50–99, four items per persona (124 items). Both full conversations are shown, headed "Conversation (date: ...)" with dates drawn so that the earlier conversation in the source gets the earlier date; a "Current date" line follows, and the model answers in one sentence.

ConvoMem-fresh. Built after all design decisions from the parts of ConvoMem (revision e3e9b391) that no earlier experiment used: two-evidence items of personas 50–99 from the fifth item of each persona onward (300 sampled from about 2,650), and three- to six-evidence items of the same personas (150 sampled per number of updates; 600 in total). Each of the k conversations contains one evidence message and is headed by an increasing date in 2024; a "Current date" line follows, and the model answers in one sentence. Chronological and newest-first orders. The judge sees the second-to-last and last evidence as old and new.

MemConflict. From the released benchmark file (30 simulated users), dynamic-conflict questions of medium difficulty whose reference answer names both the old and the new value of exactly one changed attribute. We keep a question only if the old value does not occur in the update session and does occur in an earlier session (structural filter, no model outputs used), take the latest such session, the update session and the sessions immediately before the update up to about 20k source tokens, and sample at most 8 questions per user. A few malformed messages in the source (28 of 142k) are rendered by their available text. The restated-value control uses the same construction on the questions whose update session mentions the old value.

PersonaMem. PersonaMem-v1, 32k contexts. The context up to the question is split into sessions at the persona system message; sessions receive increasing dates in source order and are shown oldest or newest first; four-way multiple choice.

LoCoMo. All 321 temporal questions (category 2) over the full conversations, each session headed by its original date and time. The official-pipeline comparison uses LoCoMo's own evaluation code (sessions newest first) against the same sessions in chronological order, 1,540 questions of categories 1–4, F1.

LongMemEval official RAG. The official retrieval and generation code with BM25 over sessions, the top five sessions per question, the official reading prompt ("I will give you several history chats ...") with the sessions in JSON, and the official grading prompt scored by our judge model; date order is the official default, relevance order skips the sort.

Retrieval order. BM25 ($k_1 = 1.5$, $b = 0.75$) between the question and each session (or fact), computed within each item; ties keep source order.

Long prompts on vLLM. Four Test of Time prompts exceed Qwen3's 40,960 positions and are skipped on vLLM; comparisons across engines use the common items.

D Decoupled answer-only training

Data. We generate 5,000 synthetic training samples (9.4M tokens) in formats that appear in no test: sticky notes, ticket updates, CSV change logs, key–value configuration snapshots, ledgers, sensor reports and meeting minutes. Each sample updates one attribute $k = 1$–6 times among distractor records and filler prose (0.3k–9.5k tokens). Records of the same attribute use templates drawn from one pool, contain no cue words ("now", "updated", "latest"), and differ only in their dates, written in mixed formats. Questions ask for the current value (60%), the earliest value (20%) or the value as of a date (20%), so that "always pick the newest date" is not a shortcut. In 20% of samples there are no dates; the answer is then the last record, or the first if the text states that it is ordered newest first. In the **decoupled** condition records are presented chronologically, reversed or shuffled (one third each), so position carries no information about time. The **control** contains the identical samples, always chronological; we verified that the two training sets differ only in record order.

Training and evaluation. We fine-tune Qwen3-4B with LoRA [20] (rank 16, $\alpha = 32$, learning rate 10^{-4}, one epoch, final checkpoint), computing the loss on answer tokens only. Hyperparameters were fixed on a development set in two further formats (calendar entries, audit tables) before any test was run; the evaluation scripts were frozen and hashed beforehand.

Results. Decoupled training transfers to formats it never saw (Table 7). On newest-first logs it reaches 100% (control 89.3%, base 70.3%). On newest-first chains of thought with step numbers it reaches 74.7% (control 2.0%, base 6.0%), although training contained dates and never step numbers. It repairs reading of the model's own lists: with a prefilled newest-to-oldest timeline and newest-first input it answers 77.1% correctly (control 51.4%; $+25.7\,[14.3, 37.1]$); when the model writes the timeline itself the gain is $+19.2\,[9.0, 29.5]$, and Phi-4-mini shows the same pattern ($+27.1\,[14.3, 40.0]$). Training only on decoupled chains of thought transfers in the other direction, to dated logs ($+28.7\,[24.7, 32.9]$ over its control), suggesting a single skill. The same pre-registered pair at Qwen3-1.7B transfers as well (newest-first logs: base 29.3%, control 1.4%, decoupled 78.3%; step-numbered newest-first CoT: 1.7%, 0.0%, 27.3%).

Limits. The pre-registered primary endpoint was not met: on newest-first LongMemEval sessions the decoupled model scores 48.7% (control 44.9%, base 51.3%). On TempReason [42], which requires matching dates to intervals, decoupled training scores $-7.0\,[-11.7, -2.7]$ points relative to the control (1.7B: -10.7), possibly an over-preference for the latest record. Because every training target was a bare value, both LoRAs also stopped following answer-format instructions: on our

memory test, which asks for one sentence, 55.1% (decoupled) and 46.4% (control) of the answers have at most three words, against 0.9% for the base model. At 4B, GSM8K [10] is unaffected (90.4% base, 92.4% both LoRAs; the models still reason step by step), but at 1.7B both LoRAs stop reasoning (median answer length 57 words before, 2 after) and GSM8K falls from 78.4% to 19.6% (decoupled) and 26.0% (control), as does BABILong's two-supporting-facts task [25] (31.3%→0.7% and 2.7%); our pre-registered limit for general ability (≤2 points) is badly missed. We therefore use this training only as evidence about the mechanism.

Table 7: Decoupled answer-only training (pre-registered; Qwen3-4B, seed 0). Accuracy (%) under non-chronological presentation; Δ = decoupled − chronological-only control with 95% paired bootstrap interval. No test format appears in training. MemoryAgentBench and LongMemEval are *seen* (§3). †Exploratory follow-up (E2), not part of the pre-registered test list.

Test	Condition	Base	Control	Decoupled	Δ
Logs (git, CHANGELOG, e-mail)	newest first	70.3	89.3	100.0	+10.7 [8.7, 12.7]
CoT, step numbers	newest first	6.0	2.0	74.7	+72.7 [67.7, 77.7]
CoT, step numbers	shuffled	34.7	27.3	45.7	+18.3 [14.0, 22.7]
Prefilled list (new→old)†	newest first	57.1	51.4	77.1	+25.7 [14.3, 37.1]
MemoryAgentBench, single hop	shuffled	42.0	57.0	70.0	+13 [4, 23]
MemoryAgentBench, multi hop	newest first	5.0	4.0	19.0	+15 [8, 23]
LongMemEval sessions	newest first	51.3	44.9	48.7	+3.8 [−3.8, 11.5]
Test of Time	shuffled	28.0	29.2	32.4	+3.1 [−2.9, 9.2]
TempReason	newest first	70.3	78.3	71.3	−7.0 [−11.7, −2.7]

E Versions of the trained procedure evaluated on held-out data

We report every version of the long-session procedure that we evaluated on a held-out set, in order, with what changed and why. Each version's data and thresholds were written before it was trained. All are Qwen3-4B LoRA (rank 16, $\alpha = 32$, learning rate 10^{-4}, one epoch, maximum length 8,192) unless noted; "thinking" results use Qwen3's thinking mode with official sampling.

- **E12** (with a chronological-only control): answer after a dated list, trained on records with local dates. ConvoMem (short, personas 50–99): newest first +2.4 over its control and +9.7 over the base model, below the threshold of +10 for both. The control learned to restate evidence in presentation order.
- **E13** (decoupled and chronological-only arms) and **E13b** (answer only): long block-dated documents, list then a bare value. ConvoMem-long newest first 95.2 and 96.0 (base 77.4); on MemConflict the bare answers did not state the change (chronological accuracy fell 9.2 and 5.4 points), and PersonaMem fell 11–18 points. Dropped because it did not follow answer formats. E13b served as the diagnostic showing that the list is needed only when dates are far from the facts (LongMemEval, seen: −16.7 [−26.9, −6.4] without the list; ConvoMem-long: −0.8).
- **E17**: same documents; varied question types and answer formats. ConvoMem-long 96.8/96.8; MemConflict 72.5/72.5/73.3 (first version designed with MemConflict's aggregate failure types); PersonaMem 60.1/57.6, because it always wrote the list, even when asked for a letter.
- **E18**: list moved into the thinking block; thinking on and off; multiple-choice, single-record and absent-attribute questions (designed with PersonaMem's aggregate failure types). Thinking off: general ability met (MMLU +0.5, ARC-Challenge +1.5, HellaSwag −0.2, IFEval +1.8, LongBench four-task average +2.3, GSM8K −1.6 [−5.6, 2.4]), MemConflict newest first 68.8

(base 36.2). Thinking on: ConvoMem-long 97.6/96.8, MemConflict 72.9/71.7/70.0, PersonaMem 63.9/56.7 (10.9% unfinished thoughts).

- **E18.1**: adds reason questions, sentence options and 11.9% self-distilled general data. Seed 0 (cloud HF): ConvoMem-long 96.8/96.8, MemConflict 80.4/77.5/77.9, PersonaMem 71.4/71.0, format threshold not met on as-of-date questions; thinking-off general ability met on the desktop (MMLU +0.6, ARC-Challenge −1.1, HellaSwag −0.3, IFEval +1.3, LongBench +3.0, GSM8K −0.8). Seed 1 (desktop HF): MemConflict 70.4/62.1/70.8. On LongMemEval's official pipeline (seen) it does not reduce the relevance-order cost (§9). *Qwen3-8B:* same recipe and self-distilled data from the 8B model; PersonaMem 71.4/67.6 against the base model's 77.3/75.6 without thinking (threshold missed); after seeing this result we added a presence penalty of 1.5 at decoding (72.7/71.0, threshold met), but the base 8B model with the same penalty alone scores 76.5/75.6; MemConflict 80.8/80.8/81.7 with the penalty against 80.4/63.7/72.5 for the base model with thinking and no penalty (two variables differ); general ability met with thinking off, on and with the penalty.
- **E19** (designed from the LongMemEval diagnosis of §9; LongMemEval is seen): adds 1,000 synthetic multi-session chats (session headers, bracketed log lines or markdown headers; never LongMemEval's JSON format), presented chronologically, reversed, shuffled or relevant-first. LongMemEval newest-ranked-first 67.6 (threshold 73.0 not met); PersonaMem 73.1/71.0; ConvoMem-long 97.6/95.2; MemConflict 70.0/70.4/70.8 (threshold missed; same machine as E18.1 seed 1, $+2.6\,[-0.7, 6.0]$ over it); general ability met (MMLU +0.8, ARC −0.7, HellaSwag −1.1, IFEval +0.4, LongBench +1.2, GSM8K +2.0). Training thereby added chat formats, so ConvoMem, MemConflict and PersonaMem are no longer unseen *formats* for E19 and later versions.
- **E19.1**: every memory-question thought starts with the dated list (absent attributes: an empty list). One-line thoughts 0%, MemConflict 78.3/77.9/78.8, LongMemEval 73.0 (list 78%), ConvoMem-long 97.6/95.2, general ability met; PersonaMem 65.5/65.5 (threshold not met; lists in 45% of its thoughts).
- **E19.2**: the 377 thinking-on reason and absent-attribute training thoughts are replaced by the base model's own thoughts on the same prompts, kept only when its visible answer was correct (335 replaced, 42 kept). MemConflict 79.2/74.2/78.3 with one-line thoughts in 4% and lists in 3%; PersonaMem, ConvoMem-long, LongMemEval and general ability not finished (the evaluation was paused after 68 of 238 PersonaMem items).

F Engines and decoding

vLLM reproduces HF item by item on 97.2% (dated logs) and 95.3% (E15) of short items and on 94.0% of 37k-token items, with condition means within 2 points; the last figure missed our pre-registered 95% criterion, so all model comparisons are made within one engine. On MemConflict, Qwen3-4B without thinking gives −40.4 on vLLM and −39.2 on HF. The thinking-mode scale study ran Qwen3-8B with one request at a time and Qwen3-14B and 32B with six concurrent requests to the vLLM server, with the same prompts, sampling, per-item seeds and parsing; batching changes numerics slightly (in a four-item check, three thinking texts differed from the one-at-a-time run). Qwen3-4B's thinking runs on MemConflict and on the LongMemEval items of §9 were made on HF on the desktop machine; its LongMemEval pipeline runs (§6) on vLLM.

G Training data details

[To be expanded: generators `gen_train_data.py` (decoupled answer-only training), `gen_bind_data_v4.py`, `gen_chat_data.py`, `gen_bind_data_lf.py`, `gen_reason_distill.py` and `gen_selfdistill.py` (trained procedure); data hashes (E18.1 train 0B46429F…, val 12CF83BB…, dev CDFADCCF…; E19.1 train D214AF41…; LF); development-set curves; the full pre-registrations with their change logs.]